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A general modified prime sieve

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ABSTRACT

Let $r \geq 2$ be an integer, q a prime power and $\mathbb{F}_q$ the finite field with q elements. Consider the problem of showing existence of primitive elements in a subset $\mathcal{A} \subseteq \mathbb{F}_{q^r}$. We prove a sieve criterion for existence of such elements, dependent only on an estimate for the character sum $\sum_{\gamma \in \mathcal{A}} \chi(\gamma)$. The flexibility and direct applicability of our criterion should be of considerable interest for problems in this field. We demonstrate the utility of our result by tackling a problem proposed by Fernandes and Reis [8] where $\mathcal{A}$ avoids a set of affine hyperplanes. This problem was studied further by Grzywaczyk and Winterhof in [9], and by Cheng and Winterhof in [4]. We obtain significant improvements to their results.

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1. Introduction

Let $\mathbb{F}_{q^r}$ be the finite field with q^r elements for some prime power q . An element $\gamma \in \mathbb{F}_{q^r}$ is called *primitive* if it generates $\mathbb{F}_{q^r}^*$, a cyclic group of order $q^r - 1$. For each divisor $e \mid q^r - 1$, call an element $\gamma \in \mathbb{F}_{q^r}^*$ *e-free* if there is no $\beta \in \mathbb{F}_{q^r}^*$ satisfying $\gamma = \beta^d$ whenever $d \mid e$ and $d \neq 1$. It follows that $\gamma \in \mathbb{F}_{q^r}^*$ is primitive if and only if it is $(q^r - 1)$ -free.

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Vinogradov provided an indicator function for when $\gamma \in \mathbb{F}_{q^r}$ is e -free, explored in more detail in [11, Lemma 2]. Let χ_d denote an arbitrary character of $\mathbb{F}_{q^r}^*$ of fixed order d . Then

$$\rho(e) \sum_{d|e} \frac{\mu(d)}{\varphi(d)} \sum_{\chi_d} \chi_d(\gamma) = \begin{cases} 1 & \gamma \text{ } e\text{-free} \\ 0 & \text{otherwise} \end{cases}, \quad (1)$$

where ρ denotes the arithmetic function $\rho : \mathbb{N} \rightarrow \mathbb{C} : m \mapsto \varphi(m)/m$. Let the quantity $N(e, \mathcal{A})$ denote the number of e -free elements in some subset $\mathcal{A} \subseteq \mathbb{F}_{q^r}$ and define the sum $S(\mathcal{A}, \chi) := \sum_{\gamma \in \mathcal{A}} \chi(\gamma)$. Suppose that, for any divisor $d \mid q^r - 1$ other than 1, we can construct an upper bound

$$|S(\mathcal{A}, \chi_d)| \leq K(q, r), \quad (2)$$

where $K(q, r)$ is independent of order d and character χ_d .

The primary goal of this paper is to provide a general criterion for the existence of a primitive element $\gamma \in \mathcal{A}$, depending only on the bound $K(q, r)$. Our criterion is proved via the *modified prime sieve*, which has already appeared in the literature. However, each proof of the criterion proceeds similarly, with some details varying depending on the application. We give a proof that applies immediately to any set $\mathcal{A}$ if we know a corresponding bound $K(q, r)$. The authors are not aware of existing results of this kind, and it is hoped that ours will save future authors from needing to prove the criterion again.

Specifically, we prove the following.

Theorem 1. (The modified prime sieve) *Consider a subset $\mathcal{A} \subseteq \mathbb{F}_{q^r}$ and assume $\text{rad}(q^r - 1) = k \prod_{i=1}^{s_1} p_i \prod_{j=1}^{s_2} l_j$ for some integer k and primes p_i, l_j . Define the quantities $\delta := \delta(p_1, \dots, p_{s_1}) = 1 - \sum_{i=1}^{s_1} \frac{1}{p_i}$ and $\epsilon := \epsilon(l_1, \dots, l_{s_2}) = \sum_{j=1}^{s_2} \frac{1}{l_j}$. Suppose $K(q, r)$ satisfies (2) and $\delta \cdot \rho(k) > \epsilon$. If*

$$|\mathcal{A}| > \frac{\rho(k)W(k)(s_1 + 2\delta - 1) + s_2 - \delta\rho(k) - \epsilon}{\delta\rho(k) - \epsilon} K(q, r), \quad (3)$$

then $\mathcal{A}$ contains a primitive element.

Here and throughout, $W(k) = 2^{\omega(k)}$ where $\omega(k)$ denotes the arithmetic function counting unique prime divisors of k . If q is a fixed prime power, we adopt the notation $\omega_r := \omega(q^r - 1)$.

As mentioned, sieve criteria are abundant in the literature for a variety of subsets $\mathcal{A} \subset \mathbb{F}_{q^r}$. Cohen and Huczynska first described the prime sieve (the special case $s_2 = 0$) in [6], and used it to prove the primitive normal basis theorem. Further examples include articles by Cohen [5, Proposition 4.3], Bailey, Cohen, Sutherland and Trudgian [2, Lemma 1] or Booker, Cohen, Leong and Trudgian [3]. The first author used the sieve combined

with a hybrid lower bound on $\omega(q^r - 1)$ to find primitive elements on lines in $\mathbb{F}_{q^r}$, see [1] for details.

We demonstrate the utility of our theorem by tackling a problem of Fernandes and Reis [8] with $\mathcal{A}$ avoiding affine hyperplanes; however, we use only the prime sieve. The authors did apply the modified prime sieve to the hyperplane problem, but found that this gave no improvement.

Let $C = \{C_1, \dots, C_r\}$ be a set of $\mathbb{F}_q$ -affine hyperplanes of $\mathbb{F}_{q^r}$ in general position and consider $\mathcal{G} = \mathbb{F}_{q^r} \setminus \bigcup_{i=1}^r C_i$. We show that $\mathcal{G}$ contains a primitive element for any choice of affine hyperplanes, for all but the following list of potential exceptional pairs (q, r) . Combining our work with [8, Theorem 1.1], we have:

Theorem 2. *For any (q, r) , $\mathcal{G}$ contains a primitive element, except possibly in the following cases:*

- $q = 13$ and $r = 4$,
- $q = 11$ and $r \in \{4, 6, 12\}$,
- $q = 9$ and $r \in \{6, 8, 9, 10, 12, 15\}$,
- $q = 8$ and $r \in \{6, 8, 10, 12, 16, 20\}$,
- $q = 7$ and $r \in \{3, 4, \dots, 12\} \cup \{14, 15, 16, 18, 20\}$,
- $q = 5$ and $r \in \{2, 3, \dots, 25\} \cup \{27, 30, 35\}$,
- $q = 4$ and $r \in \{3, 4, \dots, 33\} \cup \{35, 36, 37, 39, 41, 42, 43\} \cup \{45, 47, 49, \dots, 87\} \cup \{91, 93, 95, 99, 105, 111, 115, 117, 123, 135\}$,
- $q = 3$ and, r is odd or $r \in \{2, 4, 6, \dots, 84\} \cup \{88, 90, 92, \dots, 104\} \cup \{108, 112, 114, 120, 144\}$.

For most of these cases, we make no claim about whether these possible exceptions are genuine (i.e. $\mathcal{G}$ actually does not contain a primitive element for some choice of hyperplanes). However, the authors have verified that the cases $(q, r) = (3, 2)$, $(5, 2)$ and $(3, 3)$ are genuine exceptions.

The outline of this paper is as follows. In §2 we prove Theorem 1 in the general setting of $\mathcal{A} \subset \mathbb{F}_{q^r}$. We introduce the modified sieve and associated character sum bound in the hyperplane setting in §3. We give explicit upper bounds for ω_r in the cases where $\mathcal{A}$ might fail to have a primitive element. This is done by applying the modified sieve first in the general setting in §4, then with a specific factorisation of $q^r - 1$ in §5. Finally, we verify some genuine exceptional pairs (q, r) in §6.

2. The modified prime sieve

The goal of this section is to prove Theorem 1 by way of Lemmas 3-7.

Lemma 3. *Let χ_d denote any character of $\mathbb{F}_{q^r}$ of order d and let e be a divisor of $q^r - 1$. Then*

$$N(e, \mathcal{A}) = \rho(e) \left(|\mathcal{A}| + \sum_{1 < d|e} \frac{\mu(d)}{\varphi(d)} \sum_{\chi_d} S(\mathcal{A}, \chi_d) \right).$$

Proof. This follows from considering the indicator function in (1) for e -free elements and summing over $\gamma \in \mathcal{A}$. Since the sums are finite, we may swap the order of summation. Treating the case when $(d = 1) \mid e$ separately, we obtain the result. $\square$

Lemma 4. Suppose $e \mid q^r - 1$ and $K(q, r)$ satisfies (2). Then

$$N(e, \mathcal{A}) \geq \rho(e) (|\mathcal{A}| - (W(e) - 1)K(q, r)). \quad (4)$$

If, in addition, $d \mid e$ and $\gcd(d, e/d) = 1$, then

$$\left| N(e, \mathcal{A}) - \rho\left(\frac{e}{d}\right) N(d, \mathcal{A}) \right| \leq \rho(e) (W(e) - W(d)) K(q, r). \quad (5)$$

Proof. For (4), write $N(e, \mathcal{A})$ in the form from Lemma 3 and bound $N(e, \mathcal{A})$ from below via taking absolute values:

$$N(e, \mathcal{A}) \geq \rho(e) \left(|\mathcal{A}| - \left| \sum_{1 < d|e} \frac{\mu(d)}{\varphi(d)} \sum_{\chi_d} S(\mathcal{A}, \chi_d) \right| \right).$$

The number of squarefree divisors d of e is given by $W(e)$ and the non-vanishing of $\mu(d)$ coincides exactly with these divisors. (4) follows by repeated application of the triangle inequality and (2). For (5), write $N(e, \mathcal{A}) - \rho\left(\frac{e}{d}\right) N(d, \mathcal{A})$ in the form from Lemma 3. Since e/d and d are coprime, we have $\rho(e/d)\rho(d) = \rho(e)$. We take absolute values and apply (2), completing the proof. $\square$

Lemma 5. Let $e_1, e_2 \mid q^r - 1$, $E = \text{lcm}(e_1, e_2)$ and $e = \gcd(e_1, e_2)$. Then

$$N(E, \mathcal{A}) \geq N(e_1, \mathcal{A}) + N(e_2, \mathcal{A}) - N(e, \mathcal{A}).$$

Proof. We claim the following two statements: for any $\gamma \in \mathcal{A}$,

$$\begin{aligned} \gamma \text{ is } e_1\text{- and } e_2\text{-free} &\iff \gamma \text{ is } E\text{-free} \\ \gamma \text{ is not } e\text{-free} &\implies \gamma \text{ is neither } e_1\text{- nor } e_2\text{-free.} \end{aligned}$$

Suppose γ is e_1 - and e_2 -free. If $\gamma = \beta^d$ for some $d \mid E$, then write $d = d_1 d_2$ with $d_1 \mid e_1$ and $d_2 \mid e_2$. Then $\gamma = (\beta^{d_1})^{d_2} \implies d_2 = 1$ because γ is e_2 -free. Similarly, $d_1 = 1$, so γ is E -free. Conversely, if γ is E -free, then $\gamma = \beta^d$ for $d \mid E$ implies $d \mid e$ so $d = 1$. Similarly, γ is e_2 -free.

Now, assume that γ is not e -free, so $\gamma = \beta^d$ for some $1 < d \mid e$. Since $d \mid e_1, e_2$, γ is neither e_1 - nor e_2 -free.

Write $\mathcal{A}^d$ for the d -free elements in $\mathcal{A}$; then

$$\mathcal{A}^{e_1} \cup \mathcal{A}^{e_2} \subseteq \mathcal{A}^e \quad \text{and} \quad \mathcal{A}^{e_1} \cap \mathcal{A}^{e_2} = \mathcal{A}^E.$$

By considering cardinalities of the finite sets $\mathcal{A}^d$, we have

$$\begin{aligned} N(e_1, \mathcal{A}) + N(e_2, \mathcal{A}) &= |\mathcal{A}^{e_1} \cup \mathcal{A}^{e_2}| + |\mathcal{A}^{e_1} \cap \mathcal{A}^{e_2}| \leq |\mathcal{A}^e| + |\mathcal{A}^E| \\ &= N(e, \mathcal{A}) + N(E, \mathcal{A}). \quad \square \end{aligned}$$

Lemma 6. Let $e \mid q^r - 1$ and suppose $\text{rad}(e) = k \prod_{i=1}^s p_i$ for a choice of k . Then

$$N(e, \mathcal{A}) \geq (1-s)N(k, \mathcal{A}) + \sum_{i=1}^s N(kp_i, \mathcal{A}) \quad (6)$$

$$= \delta(p_1, \dots, p_s)N(k, \mathcal{A}) + \sum_{i=1}^s \left(N(kp_i, \mathcal{A}) - \left(1 - \frac{1}{p_i}\right) N(k, \mathcal{A}) \right). \quad (7)$$

Proof. Observe that $N(e, \mathcal{A}) = N(\text{rad}(e), \mathcal{A})$. Indeed, we have

$$\mathcal{A}^{\text{rad}(e)} \supseteq \mathcal{A}^e \cap \mathcal{A}^{\text{rad}(e)} = \mathcal{A}^{\text{lcm}(e, \text{rad}(e))} = \mathcal{A}^e.$$

Assume γ is $\text{rad}(e)$ -free and $\gamma = \beta^d$ for some $d \mid e$. Letting $e = \prod_{i=1}^s p_i^{a_i}$ and $d = \prod_{i=1}^{s'} p_i^{b_i}$ for $1 \leq b_i \leq a_i$ and $s' \leq s$, we write $\gamma = \beta^d = (\beta')^{\prod_{i=1}^{s'} p_i^{b_i}}$, say. Since γ is $\text{rad}(e)$ -free and $\prod_{i=1}^{s'} p_i \mid \text{rad}(e)$, we conclude $\gamma = \beta' = \beta^{\prod_{i=1}^{s'} p_i^{b_i-1}}$. We can repeat the argument and reduce the multiplicity of p_i in d further, until concluding that $d = 1$. This shows that γ is e -free and indeed $\mathcal{A}^e = \mathcal{A}^{\text{rad}(e)}$, as claimed. We proceed by applying Lemma 5 repeatedly:

$$\begin{aligned} N(\text{rad}(e), \mathcal{A}) &\geq N(kp_1, \mathcal{A}) + \left(N(kp_2 \cdots p_s, \mathcal{A}) - N(k, \mathcal{A}) \right) \\ &\geq N(kp_1, \mathcal{A}) + N(kp_2, \mathcal{A}) + \left(N(kp_3 \cdots p_s, \mathcal{A}) - 2 \cdot N(k, \mathcal{A}) \right) \\ &\vdots \\ &\geq \sum_{i=1}^s N(kp_i, \mathcal{A}) - (s-1)N(k, \mathcal{A}). \end{aligned}$$

The second equality is a direct rearrangement of the first. $\square$

Lemma 7. Let $e \mid q^r - 1$ and suppose $\text{rad}(e) = k \prod_{i=1}^{s_1} p_i \prod_{j=1}^{s_2} l_j$. Then

$$N(e, \mathcal{A}) \geq N(kp_1 \cdots p_{s_1}, \mathcal{A}) - \epsilon(l_1, \dots, l_{s_2})|\mathcal{A}| + \sum_{j=1}^{s_2} \left(N(l_j, \mathcal{A}) - \left(1 - \frac{1}{l_j}\right) |\mathcal{A}| \right).$$

Proof. Let $k_1 = kp_1 \dots p_{s_1}$. Applying Lemma 6, we have

$$\begin{aligned} N(e, \mathcal{A}) &\geq (1 - s_2)N(k_1, \mathcal{A}) + \sum_{j=1}^{s_2} N(k_1 l_j, \mathcal{A}) \\ &= N(kp_1 \dots p_{s_1}, \mathcal{A}) + \sum_{j=1}^{s_2} \left(N(k_1 l_j, \mathcal{A}) - N(k_1, \mathcal{A}) \right). \end{aligned}$$

It can be seen that for any $j \in [1, s_2]$, we have equality in Lemma 5 if we use $e_1 = k_1$ and $e_2 = l_j$. Using this fact, we obtain

$$\begin{aligned} N(e, \mathcal{A}) &\geq N(kp_1 \dots p_{s_1}, \mathcal{A}) + \sum_{j=1}^{s_2} \left(N(k_1 l_j, \mathcal{A}) - N(k_1, \mathcal{A}) \right) \\ &= N(kp_1 \dots p_{s_1}, \mathcal{A}) + \sum_{j=1}^{s_2} \left(N(l_j, \mathcal{A}) - N(1, \mathcal{A}) \right) \\ &\geq N(kp_1 \dots p_{s_1}, \mathcal{A}) - \epsilon(l_1, \dots, l_{s_2})|\mathcal{A}| + \sum_{j=1}^{s_2} \frac{1}{l_j} |\mathcal{A}| + \sum_{j=1}^{s_2} \left(N(l_j, \mathcal{A}) - |\mathcal{A}| \right), \end{aligned}$$

which completes the proof. $\square$

Proof of Theorem 1. Applying (5) with $e = kp_i$ and $d = k$, and noting $\rho(p_i) = 1 - p_i^{-1}$ we obtain

$$\begin{aligned} \left| N(kp_i, \mathcal{A}) - \left(1 - \frac{1}{p_i} \right) N(k, \mathcal{A}) \right| &\leq \rho(kp_i) (W(kp_i) - W(k)) K(q, r) \\ &= \rho(k) \rho(p_i) (W(k)W(p_i) - W(k)) K(q, r) \\ &= \rho(k) W(k) K(q, r) \rho(p_i). \end{aligned}$$

Summing over all $i \in [1, s_1]$, we obtain

$$\begin{aligned} \sum_{i=1}^{s_1} \left| N(kp_i, \mathcal{A}) - \left(1 - \frac{1}{p_i} \right) N(k, \mathcal{A}) \right| &\leq \sum_{i=1}^{s_1} \rho(k) W(k) K(q, r) \rho(p_i) \\ &\leq \rho(k) W(k) K(q, r) \sum_{i=1}^{s_1} \left(1 - \frac{1}{p_i} \right) \\ &= \rho(k) W(k) K(q, r) (s_1 + \delta - 1). \end{aligned}$$

By the same logic, applying (5) with $e = l_j$ and $d = 1$, before subsequently summing over all $j \in [1, s_2]$, we obtain

$$\begin{aligned}
\sum_{j=1}^{s_2} \left| N(l_j, \mathcal{A}) - \left(1 - \frac{1}{l_j}\right) |\mathcal{A}| \right| &= \sum_{j=1}^{s_2} \left| N(l_j, \mathcal{A}) - \left(1 - \frac{1}{l_j}\right) N(1, \mathcal{A}) \right| \\
&\leq \sum_{j=1}^{s_2} \rho(l_j) (W(l_j) - W(1)) K(q, r) \\
&= (s_2 - \epsilon) K(q, r).
\end{aligned}$$

Applying (4) with $e = k$, we obtain

$$N(k, \mathcal{A}) \geq \rho(k) (|\mathcal{A}| - (W(k) - 1) K(q, r)).$$

By Lemma 7 with $e = q^r - 1$, we have

$$N(q^r - 1, \mathcal{A}) \geq N(kp_1 \cdots p_{s_1}, \mathcal{A}) - \epsilon |\mathcal{A}| + \sum_{j=1}^{s_2} \left(N(l_j, \mathcal{A}) - \left(1 - \frac{1}{l_j}\right) |\mathcal{A}| \right).$$

Applying Lemma 6 to $N(kp_1 \cdots p_{s_1}, \mathcal{A})$, we arrive at

$$\begin{aligned}
N(q^r - 1, \mathcal{A}) &\geq \delta N(k, \mathcal{A}) + \sum_{i=1}^{s_1} \left(N(kp_i, \mathcal{A}) - \left(1 - \frac{1}{p_i}\right) N(k, \mathcal{A}) \right) \\
&\quad - \epsilon |\mathcal{A}| + \sum_{j=1}^{s_2} \left(N(l_j, \mathcal{A}) - \left(1 - \frac{1}{l_j}\right) |\mathcal{A}| \right).
\end{aligned}$$

We use (4) when $e = k$ and apply our bounds for the absolute value of the sums:

$$\begin{aligned}
N(q^r - 1, \mathcal{A}) &\geq \delta \rho(k) (|\mathcal{A}| - (W(k) - 1) K(q, r)) - \epsilon |\mathcal{A}| \\
&\quad - \rho(k) W(k) K(q, r) (s_1 + \delta - 1) - (s_2 - \epsilon) K(q, r) \\
&= (\delta \rho(k) - \epsilon) |\mathcal{A}| - (\rho(k) W(k) (s_1 + 2\delta - 1) + s_2 - \delta \rho(k) - \epsilon) K(q, r).
\end{aligned}$$

Thus, to ensure $N(q^r - 1, \mathcal{A}) > 0$, it is sufficient to satisfy

$$(\delta \rho(k) - \epsilon) |\mathcal{A}| > (\rho(k) W(k) (s_1 + 2\delta - 1) + s_2 - \delta \rho(k) - \epsilon) K(q, r).$$

Assuming $\delta \rho(k) > \epsilon$, and dividing through, the result follows. $\square$

For computations, it is often useful to have special cases of Theorem 1.

Corollary 8. (The prime sieve) Let q be a prime power, $r \in \mathbb{N}$ and $\mathcal{A} \subseteq \mathbb{F}_{q^r}$. Choose a core $k \in \mathbb{N}$ and $p_i \in \mathbb{P}$ such that $\text{rad}(q^r - 1) = k \prod_{i=1}^s p_i$. Suppose $K(q, r)$ satisfies (2) and $\delta > 0$. If

$$|\mathcal{A}| > \frac{W(k) (s + 2\delta - 1) - \delta}{\delta} K(q, r),$$

then $\mathcal{A}$ contains a primitive element.

Proof. Apply Theorem 1 for $s_1 = s$ and $s_2 = 0$. $\square$

Corollary 9. (The unsieved condition) Let q be a prime power, $r \in \mathbb{N}$ and $\mathcal{A} \subseteq \mathbb{F}_{q^r}$. Suppose $K(q, r)$ satisfies (2). If

$$|\mathcal{A}| > (W(q^r - 1) - 1) K(q, r),$$

then $\mathcal{A}$ contains a primitive element.

Proof. Apply Theorem 1 for $s_1 = s_2 = 0$. $\square$

In general, when applying Theorem 1 computationally, the goal is to choose a core k and primes p_i, l_j such that the right-hand side is minimised. For any fixed s_1, s_2 , we take l_j to be the s_2 largest factors of $\text{rad}(q^r - 1)$ and p_i to be the s_1 largest factors of $\text{rad}(q^r - 1) / \prod_{j=1}^{s_2} l_j$. We then iterate through the pairs (s_1, s_2) for which $\delta\rho(k) > \epsilon$, using the lowest bound possible. If the factorisation of $q^r - 1$ is known, we apply the result with explicit choices of p_i and l_j . If instead the factorisation of $q^r - 1$ is unknown, we can still improve over the unsieved bound by taking l_{s_2-j} and p_{s_1-i} to be the $(\omega_r - j)^{\text{th}}$ and $(\omega_r - s_2 - i)^{\text{th}}$ primes, respectively.

3. Avoiding hyperplanes

We now turn to a problem of finding primitive elements outside affine hyperplanes. Fernandes and Reis [8] proved the following:

Proposition 10. [8, Th. 1.1] Let $r \geq 2$. Then $\mathcal{G}$ contains a primitive element provided one of the following holds:

- i) $q \geq 16$
- ii) $q = 13$, and $r \neq 4$
- iii) $q = 11$ and $r \neq 4, 6, 12$
- iv) $q = 7, 8, 9$ and r sufficiently large.

Recently, Grzywaczyk and Winterhof [9, Cor. 3.5] proved that $\mathcal{G}$ also contains a primitive element when $q = 3, 4, 5$ and r sufficiently large. Their argument for $q = 3$ is derived from recent work by Iyer and Shparlinski [10]. In the case when r is even, Cheng and Winterhof [4] gave explicit bounds on r for $q = 3, 4, 5, 7, 8, 9$. Existing methods essentially rely on proving a character sum estimate $K(q, r)$ satisfying (2) and applying the unsieved criterion in Corollary 9. We use the same estimates, but instead apply the prime sieve in Corollary 8 to obtain substantial improvements.

Lemma 11. For any prime power q and any integer $r \geq 2$,

$$|S(\mathcal{G}, \chi)| \leq \sqrt{3}(q-1)^{r/2} q^{\lceil 3r/4 \rceil / 2}.$$

Proof. From [9, Th. 2.2], we have

$$|S(\mathcal{G}, \chi)| \leq (q-1)^{r/2} (2q^{3r/2-k} + q^k)^{1/2}$$

for any integer k between 1 and r . If we choose $k = \lceil 3r/4 \rceil$, then we get the result. $\square$

Remark 12. Actually, $\lceil 3r/4 \rceil$ is not necessarily the best choice for k (noting that we want the right hand side to be as small as possible). Instead, choose $k' \in \{1, \dots, r\}$ such that

$$2q^{3r/2-k'} + q^{k'} \leq 2q^{3r/2-k} + q^k \quad \text{for any } k \in \{1, \dots, r\},$$

and one can show by differentiating that this is either the floor or ceiling of $\frac{3r}{4} + \frac{\log 2}{2 \log q}$. When we know q and r , we can say exactly what the best choice of k is, and we will use it instead of $\lceil 3r/4 \rceil$.

Lemma 13. For any prime power q and even integer $r \geq 2$,

$$|S(\mathcal{G}, \chi)| < 2(q-1)^{3r/4} q^{r/8}.$$

Proof. See [4, Eq. 3.1]. $\square$

As a direct result of Corollary 8, we obtain a sieving criterion for primitive elements avoiding hyperplanes.

Proposition 14. (The hyperplane sieves) Let q be a prime power and r an integer not less than 2. Choose a core $k \in \mathbb{N}$ and $p_i \in \mathbb{P}$ such that $\text{rad}(q^r - 1) = k \prod_{i=1}^s p_i$. Suppose $\delta := \delta(p_1, \dots, p_s) > 0$ and

$$\frac{(q-1)^r}{q^{\lceil 3r/4 \rceil}} > 3 \left[2^{\omega_r - s} \left(\frac{s-1}{\delta} + 2 \right) - 1 \right]^2, \quad (8)$$

then $\mathcal{G}$ contains a primitive element. Further, if r is even and

$$\frac{(q-1)^{r/2}}{q^{r/4}} > 4 \left[2^{\omega_r - s} \left(\frac{s-1}{\delta} + 2 \right) - 1 \right]^2, \quad (9)$$

then $\mathcal{G}$ contains a primitive element.

Proof. Apply Corollary 8 with $\mathcal{A} = \mathcal{G}$ and $K(q, r) = \sqrt{3}(q-1)^{r/2} q^{\lceil 3r/4 \rceil / 2}$, which satisfies (2) by Lemma 11. Squaring both sides and rearranging terms, we obtain (8). We derive (9) in the same way using Lemma 13 in place of Lemma 11. $\square$

4. Explicit bounds on ω_r

For each q , we determine an integer B , such that $N(q^r - 1, \mathcal{G}) > 0$ whenever $\omega_r \geq B$. Such a bound enables us to verify (8) and (9) computationally. It is straightforward to find such a B , but when $q = 3, 4$, the naive approach yields unfeasibly large bounds. In those two cases we develop a leaping technique to reduce B to a computable range.

4.1. The case where $5 \leq q \leq 9$

Proposition 15. *Let q be a prime power and choose $r \in \mathbb{N}$. If*

$$\frac{1}{q}(p_1 p_2 \dots p_{\omega_r})^c > 3 \times 4^{\omega_r},$$

where $c = \log_q(q - 1) - \frac{3}{4}$, and p_j is the j^{th} prime, then $\mathcal{G}$ contains a primitive element. Further, if the hypothesis holds for $\omega_r = N$ and we have $p_N^c > 4$, then $\mathcal{G}$ contains a primitive element whenever $\omega_r \geq N$.

Proof. Apply Corollary 9 with $\mathcal{A} = \mathcal{G}$ and $K(q, r) = \sqrt{3}(q - 1)^{r/2} q^{\lceil 3r/4 \rceil / 2}$ to find that $\mathcal{G}$ will contain a primitive element whenever

$$(q - 1)^r > 2^{\omega_r} \sqrt{3}(q - 1)^{r/2} q^{\lceil 3r/4 \rceil / 2}.$$

Notice that we have $q^r - 1 \geq p_1 p_2 \dots p_{\omega_r}$. Therefore,

$$\begin{aligned} \mathcal{G} \text{ contains a primitive element} &\iff (q - 1)^{r/2} > 2^{\omega_r} \sqrt{3} q^{(3r/4 + 1)/2} \\ &\iff \frac{(q - 1)^{r/2}}{q^{3r/8} q^{1/2}} > 2^{\omega_r} \sqrt{3} \\ &\iff \frac{1}{q} \left(\frac{q - 1}{q^{3/4}} \right)^r > 3 \times 4^{\omega_r} \\ &\iff \frac{1}{q} (q^c)^r > 3 \times 4^{\omega_r} \\ &\iff \frac{1}{q} (p_1 p_2 \dots p_{\omega_r} + 1)^c > 3 \times 4^{\omega_r}, \end{aligned}$$

which is implied by the hypothesis. If $\frac{1}{q}(p_1 p_2 \dots p_N)^c > 3 \times 4^N$, then

$$\begin{aligned} \frac{1}{q}(p_1 p_2 \dots p_N p_{N+1})^c &= \frac{1}{q}(p_1 p_2 \dots p_N)^c (p_{N+1})^c \\ &> \frac{1}{q}(p_1 p_2 \dots p_N)^c (p_N)^c \\ &> 3 \times 4^N \times 4 = 3 \times 4^{N+1}, \end{aligned}$$

Table 1

Given q , a value N such that $\mathcal{G}$ contains a primitive element whenever $\omega_r \geq N$.

q	5	7	8	9
N	61367	1316	756	541

and the result holds by induction, because p_{N+1}^c is also greater than 4. $\square$

It is simple to verify that Proposition 15 gives, for each $q \in \{5, 7, 8, 9\}$, the values of N in Table 1.

4.2. The case where $q = 4$

Lemma 16. *When $q = 4$, the subset $\mathcal{G}$ contains a primitive element whenever $r \geq 3.07 \times 10^{19}$. Thus, if $\mathcal{G}$ does not contain a primitive element, $\omega_r < 1.31 \times 10^{18}$.*

Proof. Using Corollary 9 with $K(q, r) = \sqrt{3}(q-1)^{r/2}q^{\lceil 3r/4 \rceil/2}$ and $|\mathcal{G}| = (q-1)^r$, we argue as follows.

$$\begin{aligned} \mathcal{G} \text{ contains a primitive element} &\iff (q-1)^r > W(q^r-1)\sqrt{3}(q-1)^{r/2}q^{\lceil 3r/4 \rceil/2} \\ &\iff W(q^r-1) < \frac{(q-1)^{r/2}}{\sqrt{3}q^{\lceil 3r/4 \rceil/2}} \\ &\iff q^{\frac{1.92r}{\log \log q^r}} < \frac{(q-1)^r}{3q^{(3r/4)+1}} = \frac{1}{3q} \left(\frac{q-1}{q^{3/4}} \right)^r, \end{aligned}$$

where we have squared and used [8, Lemma 2.2] at the last step. Taking logarithms, we have

$$\begin{aligned} \mathcal{G} \text{ contains a primitive element} &\iff \frac{1.92r \log q}{\log r + \log \log q} < -\log(3q) + r \log \frac{q-1}{q^{3/4}} \\ &\iff \frac{1.92 \log q}{\log r + \log \log q} < -\frac{\log(3q)}{r} + \log \frac{q-1}{q^{3/4}}. \end{aligned}$$

With respect to r , the left hand side is decreasing and the right hand side is increasing. If the inequality holds for some r_0 , it holds for any $r \geq r_0$. It is easily verified, then, that the inequality holds for every $r \geq 3.07 \times 10^{19}$. If $\mathcal{G}$ does not contain a primitive element, then $r < 3.07 \times 10^{19}$. Using [8, Lemma 2.2] again, we find that

$$\begin{aligned} \omega_r = \log_2 W(q^r-1) &< \log_2 \left(q^{\frac{0.96r}{\log r + \log \log q}} \right) \\ &= \frac{0.96r \log_2 q}{\log r + \log \log q} < 1.31 \times 10^{18}, \end{aligned}$$

where we have again used the fact that $\frac{0.96r \log_2 q}{\log r + \log \log q}$ is increasing with respect to r . $\square$

Remark 17. This bound on ω_r is too large for us to apply (8) to, so we must reduce it further. Our strategy in the next proposition (which is also useful if $q > 4$) is to fix $\omega \in \mathbb{N}$ and eliminate any r with $\omega_r = \omega$. In particular, if we also fix $0 \leq t \leq \omega$, we determine an upper bound for r , such that (8) fails. By [8, Lemma 2.2], this gives an upper bound on ω_r . If this forces $\omega_r < \omega$, we conclude $N(q^r - 1, \mathcal{G}) > 0$ for all such r .

Proposition 18. *Let $q \geq 4$ be a prime power and $r \in \mathbb{N}$. Fix ω and $0 \leq t \leq \omega$, and let $s(\omega) = \omega - t$. Let*

$$\delta(\omega) = 1 - \sum_{i=\omega-s+1}^{\omega} \frac{1}{p_i} = 1 - \sum_{i=t+1}^{\omega} \frac{1}{p_i},$$

with p_i the i^{th} prime, and assume $\delta > 0$. Let

$$\begin{aligned} R(\omega) &= \log \left(3 \left[2^t \left(\frac{s(\omega) - 1}{\delta(\omega)} + 2 \right) - 1 \right]^2 \right), \\ K(R) &= \frac{R + \log q}{\log(q - 1) - \frac{3}{4} \log q}, \\ B(K) &= \frac{0.96K \log_2 q}{\log K + \log \log q}. \end{aligned}$$

If $\omega \geq (B \circ K \circ R)(\omega)$, then $\mathcal{G}$ contains a primitive element. Additionally, if we fix t but not ω , and suppose $\omega \geq t$, then $(K \circ R)$ and R are strictly increasing functions of ω .

Proof. First, note that $R(\omega)$ is the logarithm of the right hand side in Proposition 14, except that we have (possibly) changed the primes in the sum constituting δ . Fix t and ω , and suppose that we will always take the largest $s(\omega)$ primes in the factorisation of $\text{rad}(q^r - 1)$, denoting these by $p'_1, p'_2, \dots, p'_{s(\omega)}$. We have that

$$\delta(\omega) = 1 - \sum_{i=\omega-s(\omega)+1}^{\omega} \frac{1}{p_i} \leq 1 - \sum_{i=1}^{s(\omega)} \frac{1}{p'_i} := \delta'(\omega),$$

from which it follows that

$$\left(3 \left[2^t \left(\frac{s(\omega) - 1}{\delta(\omega)} + 2 \right) - 1 \right]^2 \right) \geq \left(3 \left[2^t \left(\frac{s(\omega) - 1}{\delta'(\omega)} + 2 \right) - 1 \right]^2 \right).$$

If $\frac{(q-1)^r}{q^{\lceil 3r/4 \rceil}}$ is greater than the left hand side, then it is also greater than the right hand side, and by Proposition 14 there must then be a primitive element in $\mathcal{G}$. It is now sufficient to show that $\log\left(\frac{(q-1)^r}{q^{\lceil 3r/4 \rceil}}\right) > R(\omega)$, which will happen if

$$r \log(q-1) - \left(\frac{3r}{4} + 1\right) \log q > R(\omega),$$

or equivalently,

$$r > \frac{R(\omega) + \log q}{\log(q-1) - \frac{3}{4} \log q} = (K \circ R)(\omega). \quad (10)$$

Note that this equivalence requires the denominator in (10) to be positive, which holds because $q \geq 4$. For a contradiction, suppose there is no primitive element in $\mathcal{G}$. Then $r \leq (K \circ R)(\omega)$. By [8, Lemma 2.2],

$$\begin{aligned} \omega = \log_2 W(q^r - 1) &< \frac{0.96}{\log \log(q^r)} \log_2(q^r) \\ &= \frac{0.96r \log_2(q)}{\log r + \log \log q}. \end{aligned}$$

This bound is effectively a Robin bound for $\omega(n)$, see [13] for more details. It is easy to show that this function is increasing on r as long as $r > \frac{e}{\log q}$ and $q > 1$. Since $r \geq 2$ and $q \geq 4$, these inequalities hold. It follows that

$$\omega < \frac{0.96r \log_2(q)}{\log r + \log \log q} \leq \frac{0.96(K \circ R)(\omega) \log_2(q)}{\log(K \circ R)(\omega) + \log \log q} = (B \circ K \circ R)(\omega).$$

By assumption, this is a contradiction. This completes the proof that $\mathcal{G}$ contains a primitive element. Finally, it is clear that if R is strictly increasing with respect to ω , then so is $K \circ R$. Clearly $s(\omega) = \omega - t$ strictly increases with ω , so we just need to show that δ is strictly decreasing with respect to ω . But it is clear that

$$1 - \sum_{i=t+1}^{\omega+1} \frac{1}{p_i} = 1 - \frac{1}{p_{\omega+1}} - \sum_{i=t+1}^{\omega} \frac{1}{p_i} < 1 - \sum_{i=t+1}^{\omega} \frac{1}{p_i},$$

so we are done. $\square$

Remark 19. We describe our leaping technique in the next lemma. Essentially, the strategy is to assume a lower bound on δ , then show that if the criterion in Proposition 18 holds for ω , then it holds for $\omega + 1$. By induction, the criterion holds for as long as the lower bound on δ does. This allows us to apply Proposition 18 for a large number of values of ω , all at once.

Lemma 20. *Let $q = 4$, select some ω_0 and fix an integer t between 0 and ω_0 . Set $\delta(\omega) = 1 - \sum_{i=t+1}^{\omega} (1/p_i)$ and suppose that we have $\delta(\omega_0) > m$ for some $0 < m$. Let $s(\omega) = \omega - t$, and*

$$r(\omega) = \log \left(3 \left[2^t \left(\frac{s(\omega) - 1}{m} + 2 \right) \right]^2 \right),$$

$$K(R) = \frac{R + \log q}{\log(q - 1) - \frac{3}{4} \log q},$$

$$B(K) = \frac{0.96K \log_2 q}{\log K + \log \log q},$$

where we have replaced the δ in Proposition 18 by its lower bound m , and dropped the -1 . Suppose that $\omega_0 \geq (B \circ K \circ r)(\omega_0)$, $s(\omega_0) \geq 19$, $(K \circ r)(\omega_0) \geq 6$ and $(K \circ R)(\omega_0) \geq 2$. If we find that $\delta(\omega_0 + 1) > m$, then there is a primitive element in $\mathcal{G}$ for $\omega = \omega_0 + 1$. Further, if $\delta(\omega_1) > m$ for some $\omega_1 > \omega_0$ then there is a primitive element in $\mathcal{G}$ whenever ω is between $\omega_0 + 1$ and ω_1 .

Proof. We know that $\omega_0 \geq (B \circ K \circ r)(\omega_0)$, and we will now estimate $(B \circ K \circ r)(\omega_0 + 1)$. First, we examine $r(\omega_0 + 1)$. We have

$$\begin{aligned} r(\omega_0 + 1) - r(\omega_0) &= 2 \left(\log 2^t + \log \left(\frac{s(\omega_0 + 1) - 1}{m} + 2 \right) \right) \\ &\quad - 2 \left(\log 2^t + \log \left(\frac{s(\omega_0) - 1}{m} + 2 \right) \right) \\ &= 2 \left(\log \left(\frac{s(\omega_0 + 1) - 1}{m} + 2 \right) - \log \left(\frac{s(\omega_0) - 1}{m} + 2 \right) \right) \\ &= 2 \left(\log \left(\frac{\omega_0 - t}{m} + 2 \right) - \log \left(\frac{\omega_0 - t - 1}{m} + 2 \right) \right). \end{aligned}$$

The argument of the first log exceeds the argument of the second by $\frac{1}{m}$. log is increasing, its second derivative is always negative and its gradient at x is $\frac{1}{x}$, so the difference in the last line cannot exceed

$$\frac{1}{m} \times \frac{1}{\frac{\omega_0 - t - 1}{m} + 2}.$$

We then have

$$\begin{aligned} r(\omega_0 + 1) - r(\omega_0) &\leq \frac{2}{m \frac{\omega_0 - t - 1}{m} + 2} = \frac{2}{\omega_0 - t - 1 + 2m} \\ &\leq \frac{2}{s(\omega_0) - 1} \leq \frac{1}{9}. \end{aligned}$$

It should also be clear that $r(\omega_0 + 1) - r(\omega_0) > 0$. It follows immediately that

$$0 < (K \circ r)(\omega_0 + 1) - (K \circ r)(\omega_0) \leq \frac{1/9}{\log(q - 1) - \frac{3}{4} \log q} < 2$$

by substituting $q = 4$. We now examine the properties of the function B . We have

$$\begin{aligned} B(K) &= \frac{1.92K}{\log K + \log \log 4} \text{ since } \log_2 4 = 2, \\ B'(K) &= \frac{1.92}{\log K + \log \log 4} \left(1 - \frac{1}{\log K + \log \log 4} \right), \text{ and} \\ B''(K) &= \frac{1.92}{K(\log K + \log \log 4)^2} \left(\frac{2}{\log K + \log \log 4} - 1 \right). \end{aligned}$$

It follows that $B(K)$ is strictly increasing if $\log K + \log \log 4 > 1$, i.e. $K > \frac{e}{\ln 4} \approx 1.961$. $B(K)$ is concave down if $\log K + \log \log 4 > 2$, i.e. $K > \frac{e^2}{\ln 4} \approx 5.330$. When $K \geq 6$, we have $0 < B'(K) \leq B'(6)$ by concavity and the fact that B is increasing. By evaluating $B'(6)$ we have $0 < B'(K) < \frac{1}{2}$, valid for $K \geq 6$.

Thus, when moving from ω_0 to $\omega_0 + 1$, $K \circ r$ increases by less than 2, so $(B \circ K \circ r)$ increases by less than 1 (as the gradient of B is less than $\frac{1}{2}$ if $(K \circ r)(\omega_0)$ is at least 6, which it is by assumption). That is, $0 < (B \circ K \circ r)(\omega_0 + 1) - (B \circ K \circ r)(\omega_0) < 1$. Using this, we have

$$\omega_0 + 1 \geq (B \circ K \circ r)(\omega_0) + 1 > (B \circ K \circ r)(\omega_0 + 1). \quad (11)$$

Notice also that

$$(K \circ r)(\omega_0 + 1) > (K \circ r)(\omega_0) \geq 6, \quad (12)$$

$$s(\omega_0 + 1) = \omega_0 - t + 1 > \omega_0 - t = s(\omega_0) \geq 19, \text{ and} \quad (13)$$

$$(K \circ R)(\omega_0 + 1) > (K \circ R)(\omega_0) \geq 2, \quad (14)$$

by Proposition 18 and the assumed lower bound on $(K \circ r)(\omega_0)$.

Next, we have to show that $(B \circ K \circ r)(\omega_0 + 1) > (B \circ K \circ R)(\omega_0 + 1)$. Since $\delta(\omega_0 + 1) > m$, we immediately get $r(\omega_0 + 1) > R(\omega_0 + 1)$, which implies

$$(K \circ r)(\omega_0 + 1) > (K \circ R)(\omega_0 + 1) > (K \circ R)(\omega_0) \geq 2.$$

Since B is strictly increasing with respect to K when $K \geq 2$, we have

$$(B \circ K \circ r)(\omega_0 + 1) > (B \circ K \circ R)(\omega_0 + 1).$$

It follows that $\omega_0 + 1 > (B \circ K \circ R)(\omega_0 + 1)$, so there is a primitive element in $\mathcal{G}_A$ for $\omega = \omega_0 + 1$. The final claim follows by induction, taking note of Equations (11), (12), (13), and (14). $\square$

Proposition 21. [12, Th. 2] For any $n \geq 2$, we have

$$\left| \sum_{p \leq n} \frac{1}{p} - \log \log n - M \right| \leq \frac{4}{\log(n+1)} + \frac{2}{n \log n}$$

where $0.26149 < M < 0.26150$ is the Meissel-Mertens constant.

Proposition 22. [14, Th. 1] For any $x > 1$,

$$\pi(x) < \frac{x}{\log x} \left(1 + \frac{3}{2 \log x} \right). \quad (15)$$

For any $x \geq 59$,

$$\pi(x) > \frac{x}{\log x} \left(1 + \frac{1}{2 \log x} \right). \quad (16)$$

Both Propositions 21 and 22 have since been improved, but the bounds stated here are sharp enough for our applications.

Proposition 23. Let $q = 4$. If $\mathcal{G}$ does not contain a primitive element, then $\omega_r \leq 10^5$.

Proof. The proof will follow from Lemma 20. We will show that, when $q = 4$, $\mathcal{G}$ contains a primitive element whenever $\omega_r > 10^5$. Let $\omega_0 = 10^5$, $m = 0.05$, $t = 3 \times 10^4$. Then $\delta(\omega_0) \approx 0.9024 > 0.05 = m$. We find that

$$r(\omega_0) \approx 41618.2, \quad (K \circ r)(\omega_0) \approx 706716.7 \geq 6 \text{ and } (B \circ K \circ r)(\omega_0) \approx 98361.3 \leq \omega_0.$$

Additionally, $s(\omega_0) = 10^5 - 3 \times 10^4 = 7 \times 10^4 \geq 19$. The hypotheses in Lemma 20 are satisfied since $(K \circ R)(\omega_0) \approx 706694.0 \geq 2$. If we have $\delta(\omega_1) > m$, for some $\omega_1 > 10^5$, then there is a primitive element in $\mathcal{G}$ for any ω between $10^5 + 1$ and ω_1 . We now want to know how large ω_1 can be before this hypothesis fails. We require that $1 - \sum_{i=3 \times 10^4}^{\omega_1} (1/p_i) > 0.05$, which happens if

$$\sum_{i=1}^{\omega_1} \frac{1}{p_i} < 0.95 + \sum_{i=1}^{3 \times 10^4} \frac{1}{p_i}.$$

The right hand side is more than 3.758. Using Proposition 21 with n equal to the ω_1^{th} prime, we get

$$\sum_{i=1}^{\omega_1} \frac{1}{p_i} = \sum_{p \leq n} \frac{1}{p} \leq \log \log n + M + \frac{4}{\log(n+1)} + \frac{2}{n \log n}, \quad (17)$$

and it is easy to verify that when $n = 2.93 \times 10^{12}$, the right hand side is less than 3.758. Using (16), $\pi(2.93 \times 10^{12}) > 1.038 \times 10^{11}$, which is to say that the $(1.038 \times 10^{11})^{\text{th}}$ prime is less than 2.93×10^{12} . Taking $w_1 = 1.038 \times 10^{11}$ is therefore valid, and we have shown that $\mathcal{G}$ contains a primitive element for any ω satisfying $10^5 + 1 \leq \omega \leq 1.038 \times 10^{11}$.

We now perform another “jump”. Let $\omega_0 = 1.038 \times 10^{11}$, $m = 0.05$, $t = 1.5 \times 10^7$. $\delta(\omega_0)$ is too large to calculate exactly, so we bound it using Proposition 21. We have

$$\delta(1.038 \times 10^{11}) = 1 - \sum_{i=1.5 \times 10^7+1}^{1.038 \times 10^{11}} (1/p_i) = 1 - \sum_{i=1}^{1.038 \times 10^{11}} (1/p_i) + \sum_{i=1}^{1.5 \times 10^7} (1/p_i).$$

The first sum does not exceed 3.758, using the bound from the previous step. The second sum is more than 3.228. It follows that

$$\delta(1.038 \times 10^{11}) > 1 - 3.758 + 3.228 = 0.47 > m,$$

as needed. Also,

$$\begin{aligned} r(\omega_0) &\approx 2.08 \times 10^7 \\ (K \circ r)(\omega_0) &\approx 3.53 \times 10^8 \geq 6 \\ (B \circ K \circ r)(\omega_0) &\approx 3.39 \times 10^7 \leq \omega_0 \\ s(\omega_0) &= 1.038 \times 10^{11} - 1.5 \times 10^7 \geq 19. \end{aligned}$$

Finally, since $\delta(\omega_0) < 1$, we find that $R(\omega_0) > \log(3[2^t(\frac{s(\omega_0)-1}{1}+2)-1]^2) \approx 2.08 \times 10^7$, so $(K \circ R)(\omega_0) > 3.53 \times 10^8 > 2$. Now, we have satisfied all of the hypotheses in Lemma 20, and we seek a large value of ω_1 satisfying $\delta(\omega_1) > m$. As before, we need

$$\sum_{i=1}^{\omega_1} \frac{1}{p_i} < 0.95 + \sum_{i=1}^{1.5 \times 10^7} \frac{1}{p_i}.$$

The right hand side is more than 4.1785. Using Proposition 21 with n equal to the ω_1^{th} prime, we again get (17), whose right hand side is less than 4.1785 if $n = 1.019 \times 10^{20}$. By (16), the $(2.2358 \times 10^{18})^{\text{th}}$ prime is less than 1.019×10^{20} , so it is valid to pick $\omega_1 = 2.2358 \times 10^{18}$. Therefore, $\mathcal{G}$ contains a primitive element for any ω satisfying $1.038 \times 10^{11} + 1 \leq \omega \leq 2.2358 \times 10^{18}$. Combining this with the result of the previous step and Lemma 16, we obtain the desired result. $\square$

4.3. The case where $q = 3$

We now turn our attention to the $q = 3$ case. We cannot apply (8), because the left hand side does not grow with r , owing to the fact that $\frac{(q-1)}{q^{3/4}} < 1$. Thus, we apply Criterion (9) instead. We will now present results analogous to Lemma 16, Proposition 18

and Lemma 20, and use a similar “jumping” argument to eliminate all but finitely many values of ω . In particular, we construct an explicit upper bound on ω_r above which we win, in the sense that $\mathcal{G}$ must contain a primitive element. For an explanation of the strategy of Proposition 25 (resp. Lemma 26), see Remark 17 (resp. Remark 19).

Lemma 24. *Let $q = 3$ and suppose that r is even. Then $\mathcal{G}$ contains a primitive element whenever $r \geq 4.972 \times 10^{12}$. Thus, if $\mathcal{G}$ does not contain a primitive element, $\omega_r < 2.58 \times 10^{11}$.*

Proof. Use the unsieved bound of Corollary 9 with $K(q, r) = 2(q-1)^{3r/4}q^{r/8}$ from Lemma 13. Then

$$\begin{aligned} N(q^r - 1, \mathcal{G}) > 0 &\iff q^{\frac{0.96r}{\log \log q^r}} < \frac{(q-1)^{r/4}}{2q^{r/8}} \\ &\iff \frac{0.96r \log q}{\log r + \log \log q} < -\log 2 + \frac{1}{4}r \log(q-1) - \frac{1}{8}r \log q \\ &\iff \frac{0.96 \log q}{\log r + \log \log q} < \frac{-\log 2}{r} + \frac{1}{4} \log(q-1) - \frac{1}{8} \log q. \end{aligned}$$

This holds if $r \geq 4.972 \times 10^{12}$. If $\mathcal{G}$ does not contain a primitive element, then $r < 4.972 \times 10^{12}$, and using [8, Lemma 2.2.3] we find that $\omega_r < 2.58 \times 10^{11}$. $\square$

Proposition 25. *Let $q = 3$ and let r be an even positive integer that is at least 4. Fix ω and $0 \leq t \leq \omega$, and let $s(\omega) = \omega - t$. Let*

$$\delta(\omega) = 1 - \sum_{i=\omega-s+1}^{\omega} \frac{1}{p_i} = 1 - \sum_{i=t+1}^{\omega} \frac{1}{p_i},$$

with p_i the i^{th} prime, and assume $\delta > 0$. Let

$$\begin{aligned} R(\omega) &= \log \left(4 \left[2^t \left(\frac{s(\omega) - 1}{\delta(\omega)} + 2 \right) - 1 \right]^2 \right), \\ K(R) &= \frac{R}{\frac{1}{2} \log(q-1) - \frac{1}{4} \log q}, \\ B(K) &= \frac{0.96K \log_2 q}{\log K + \log \log q}. \end{aligned}$$

If $\omega \geq (B \circ K \circ R)(\omega)$, then $\mathcal{G}$ contains a primitive element. Additionally, if we fix t but not ω , and suppose $\omega \geq t$, then $(K \circ R)$ and R are strictly increasing functions of ω .

Proof. The proof is analogous to that of Proposition 18. Note the change from 3 to 4 in the definition of $R(\omega)$, and the change in definition of $K(R)$. This occurs because we

replace $\frac{(q-1)^r}{3q^{(3r/4)+1}}$ with $\frac{(q-1)^{r/2}}{4q^{r/4}}$. Finally, recall that it was necessary to have $r > \frac{e}{\log q}$, which is the reason we exclude the case $r = 2$. $\square$

Lemma 26. *Let $q = 3$, select some ω_0 , suppose r is an even positive integer not less than 4, and fix a t between 0 and ω_0 . Let $\delta(\omega) := 1 - \sum_{i=t+1}^{\omega} (1/p_i)$ and suppose that we have $\delta(\omega_0) > m$ for some $0 < m$. Let $s(\omega) = \omega - t$, and*

$$r(\omega) = \log \left(4 \left[2^t \left(\frac{s(\omega) - 1}{m} + 2 \right) \right]^2 \right),$$

$$K(R) = \frac{R}{\frac{1}{2} \log(q-1) - \frac{1}{4} \log q},$$

$$B(K) = \frac{0.96K \log_2 q}{\log K + \log \log q}.$$

Suppose that $\omega_0 \geq (B \circ K \circ r)(\omega_0)$, $s(\omega_0) \geq 15$, $(K \circ r)(\omega_0) \geq 7$ and $(K \circ R)(\omega_0) \geq 3$. If we find that $\delta(\omega_0 + 1) > m$, then $\omega_0 + 1 \geq (B \circ K \circ r)(\omega_0 + 1) > (B \circ K \circ R)(\omega_0 + 1)$, which implies the existence of a primitive element in $\mathcal{G}$ for $\omega = \omega_0 + 1$. Additionally, δ is a decreasing function of ω , so by induction, if $\delta(\omega_1) > m$ for some $\omega_1 > \omega_0$ then there is a primitive element in $\mathcal{G}$ whenever ω is between $\omega_0 + 1$ and ω_1 .

Proof. The proof is, *mutatis mutandis*, the same as that of Lemma 20. Note that the required lower bounds on s , $(K \circ r)$ and $(K \circ R)$ have changed. The bound on s can decrease to 15 because it is now sufficient to have $r(\omega_0 + 1) - r(\omega_0) \leq \frac{1}{7}$. The bounds on $(K \circ r)$ and $(K \circ R)$ change because $B(K)$ is now strictly increasing if $K > \frac{e}{\ln 3} \approx 2.474$ and is concave down if $K > \frac{e^2}{\ln 3} \approx 6.726$. $\square$

Proposition 27. *Let $q = 3$ and suppose r is an even integer greater than 2. If $\mathcal{G}$ does not contain a primitive element, then $\omega_r \leq 10^5$.*

Proof. We perform the “jumps” as before; in fact, one jump is enough. Let $\omega_0 = 10^5$, $m = 0.01$ and $t = 3 \times 10^4$. Then $\delta(\omega_0) \approx 0.9024 > m$. We have

$$\begin{aligned} r(\omega_0) &\approx 41621.7 \\ s(\omega_0) &= 7 \times 10^4 \geq 15 \\ (K \circ r)(\omega_0) &\approx 578718.6 \geq 7 \\ R(\omega_0) &\approx 41612.5 \\ (K \circ R)(\omega_0) &\approx 578590.5 \geq 3 \\ \omega_0 &\geq (B \circ K \circ r)(\omega_0) \approx 65897.1. \end{aligned}$$

We need to find ω_1 such that $\delta(\omega_1) > m$, that is,

Table 2
The result of applying Propositions 18 and 25.

q	Values of ω eliminated	Values of r eliminated
9	$\omega \geq 27$	$r \geq 39$
8	$\omega \geq 28$	$r \geq 44$
7	$\omega \geq 31$	$r \geq 52$
5	$\omega \geq 46$	$r \geq 104$
4	$\omega \geq 120$	$r \geq 391$
3	$\omega \geq 73$	Even $r \geq 276$

$$\sum_{i=1}^{\omega_1} (1/p_i) < 0.99 + \sum_{i=1}^t (1/p_i).$$

The right hand side is smaller than 3.798. Letting n be the ω_1^{th} prime, we find that the value $n = 1.143 \times 10^{13}$ satisfies (17). Then $\pi(n) > 3.864 \times 10^{11}$, so it is valid to pick $\omega_1 = 3.864 \times 10^{11}$. It follows that $\mathcal{G}$ contains a primitive element whenever r is even and $10^5 + 1 \leq \omega(3^r - 1) \leq 3.864 \times 10^{11}$. This, with Lemma 24, proves the result. $\square$

5. Checking small values of ω and r

In what follows, we fix a value of q , then say that a value ω has been eliminated if we have proved $\mathcal{G}_A$ contains a primitive element for any r such that $\omega(q^r - 1) = \omega$. For a particular q , we now check values of $\omega < N$ using Propositions 18 and 25. For each ω , we must pick a value for t . In practice, one can usually use the same value of t to eliminate many values of ω .

For instance, when $q = 5$, the value $t = 33$ can be used to eliminate all values of ω in the range $[116, 61366]$. Putting $t = 4$ is enough to show the same for $[46, 115]$. The value $t = 3$ does not eliminate any smaller values of ω , so we stop there. Using the result in Table 1, we have eliminated every integer value of ω greater than 45. However, we are actually interested in eliminating values of r . Referring to the proof of Proposition 18 (or Proposition 25 if $q = 3$), we fix t , and pick $\omega = \omega_0 \geq t$. Then the condition $r > (K \circ R)(\omega_0)$ is enough to guarantee that $\mathcal{G}$ will contain a primitive element. But $(K \circ R)$ is an increasing function of ω , so

$$r > (K \circ R)(\omega_0) \text{ implies } r > (K \circ R)(\omega)$$

for any ω such that $t \leq \omega \leq \omega_0$. Returning to our example of $q = 5$, we select $t = 4$ and $\omega_0 = 45$. Then $r > (K \circ R)(45) \approx 103.639$ will guarantee that $\mathcal{G}$ contains a primitive element. If $\mathcal{G}$ has no primitive element, then we have $\omega \leq 45$, but when $4 \leq \omega \leq 45$ we can also be sure that $r \leq 103$. Repeating with smaller values of t , we can obtain upper bounds that are valid when ω is as small as 1.

Table 2 shows the results of these computations for other values of q . The remaining values of r are small enough that it is feasible to find and use the actual prime factorisation of $q^r - 1$.

We now employ Proposition 14 directly. We fix a pair (q, r) that we need to check, then find the distinct prime factors of $q^r - 1$ (of which there are ω_r). For each s such that $1 \leq s \leq \omega_r$, we take the largest s prime factors and use them as the p_i . We then check whether the relevant inequality in Proposition 14 holds. Note that, if we use (8), we will modify it as in Remark 12. If the (possibly modified) inequality holds for any s , we conclude that $\mathcal{G}$ contains a primitive element. Tables 3 and 4 show the pairs (q, r) that are eliminated via (8) and (9), respectively. For $q \geq 5$, Mathematica readily determined the factorisation of each $q^r - 1$ that was tested. For $q \in \{3, 4\}$, the authors resorted to the online database [7]. For each required value of r , the database provided the list of prime factors of $q^r - 1$. Having eliminated many values of r , we now apply the following two criteria (used by Fernandes and Reis in [8]) to the remaining cases. The first criterion is simple and can usually eliminate many small values of r if $q \geq 7$.

Proposition 28. *If $(q - 1)^r > q^r - \varphi(q^r - 1)$, then $\mathcal{G}$ contains a primitive element.*

Proof. The result follows from the fact that there are $\varphi(q^r - 1)$ primitive elements in $\mathbb{F}_{q^r}^*$ and $|\mathbb{F}_{q^r}| = q^r$. Indeed, if $\mathcal{G}$ contains no primitive elements, then we must have $|\mathbb{F}_{q^r} \setminus \mathcal{G}| \geq \varphi(q^r - 1)$, a contradiction. $\square$

The second criterion (proved in [8]) eliminates some larger values of r if $q \geq 7$. It involves a bound on character sums over an $\mathbb{F}_q$ -affine subspace $\mathcal{A} \subseteq \mathbb{F}_{q^n}$; the bound depends on the dimension of the subspace.

Proposition 29. [8, Th. 3.2] *If $(q - 1)^r > \alpha(q, r)2^{\omega_r}$, where*

$$\alpha(q, r) = \sum_{i=0}^{r-1} \binom{r}{i} q^{\min\{i, r/2\}},$$

then $\mathcal{G}$ contains a primitive element.

For specific q and r , it is easy to check whether either of these criteria is satisfied. Given q , we allow r to range from 2 to the largest value not eliminated in the previous step. Table 5 shows the results of applying the last two criteria. This also completes the proof of Theorem 2.

6. Some genuine exceptions

In Section 1, we observed that the possible exceptions listed in Theorem 2 might not all be genuine. We now prove that three of those possible exceptions are genuine. Specifically, we have the following result.

Table 3
The pairs (q, r) that were eliminated using (8), making use of the actual factorisation of $q^r - 1$.

q	Values of r that were checked	Values of r that were eliminated
9	$2 \leq r \leq 38$	13, 16, 17 and 19-38
8	$2 \leq r \leq 43$	13, 15, 17, 18, 19 and 21-43
7	$2 \leq r \leq 51$	17, 19, 21, 22, 23 and 25-51
5	$2 \leq r \leq 103$	29, 31, 33, 37, 39, 41, 43, 45, 46, 47 and 49-103
4	$2 \leq r \leq 390$	89, 97, 101, 103, 107, 109, 113, 119, 121, 125, 127, 129, 131, 133, 137, 139, 141-143, 145-149, 151-153, 155, 157-161, 163-167, 169-179, 181-209, 211-390

Table 4
The pairs (q, r) that were eliminated using (9), making use of the actual factorisation of $q^r - 1$.

q	Values of r that were checked (even values only)	Values of r that were eliminated
9	$2 \leq r \leq 18$	14, 16, 18
8	$2 \leq r \leq 20$	14
7	$2 \leq r \leq 24$	22, 24
5	$2 \leq r \leq 48$	26, 28, all even numbers between 32 and 48
4	$2 \leq r \leq 210$	34, 38, 40, all even numbers between 44 and 210
3	$4 \leq r \leq 268$	86, 106, 110, 116, 118, all even numbers between 122 and 268 except 144

Table 5
The pairs (q, r) that were eliminated using Propositions 28 and 29.

q	Values of r that were checked	Values of r that were eliminated
9	$2 \leq r \leq 15$	$r = 2, 3, 4, 5, 7, 11, 13, 14$
8	$2 \leq r \leq 20$	$r = 2, 3, 4, 5, 7, 9, 11, 13, 17, 18, 19$
7	$2 \leq r \leq 20$	$r = 2, 13, 19$
5	$2 \leq r \leq 35$	None
4	$2 \leq r \leq 135$	$r = 2$
3	$2 \leq r \leq 120$ (even r only), and 144	None

Proposition 30. When (q, r) is either $(3, 2)$, $(5, 2)$ or $(3, 3)$, it is possible to select hyperplanes $C_1, \dots, C_r$ such that every primitive element lies on at least one of the C_i . That is, such that $\mathcal{G}_A$ does **not** contain a primitive element.

Proof. For $(q, r) = (3, 2)$, we have the finite field extension $\mathbb{F}_{3^2}/\mathbb{F}_3$. To describe it we use the isomorphism $\mathbb{F}_9 \cong \mathbb{F}_3[x]/(x^2 - 2)$. The unit group has order $3^2 - 1 = 8$, so there are $\varphi(8) = 4$ primitive elements.

One can verify that $\alpha = 1 + x$ is a primitive element (it is sufficient to check that $\alpha^4 \neq 1$), so the set of primitive elements is

$$\{\alpha = 1 + x, \alpha^3 = 1 + 2x, \alpha^5 = 2 + 2x, \alpha^7 = 2 + x\}.$$

Select the hyperplanes

$$H_1 = \{a_1(1+x) \mid a_1 \in \mathbb{F}_3\}$$

$$H_2 = \{b_1(1+2x) \mid b_1 \in \mathbb{F}_3\}.$$

It is clear that α and α^5 lie on H_1 , and that α^3 and α^7 lie on H_2 . The set of elements not on the hyperplanes, $\mathcal{G}$, thus contains no primitive elements.

For $(q, r) = (5, 2)$, we use $\mathbb{F}_{25} \cong \mathbb{F}_5[x]/(x^2 - 2)$. $\alpha = 2 + x$ is a primitive element, so the set of primitive elements is

$$\{2 + x, 2 + 4x, 4 + 2x, 4 + 3x, 3 + 4x, 3 + x, 1 + 3x, 1 + 2x\}.$$

We choose hyperplanes

$$H_1 = \{a_1(2+x) \mid a_1 \in \mathbb{F}_5\}$$

$$H_2 = \{b_1(3+x) \mid b_1 \in \mathbb{F}_5\};$$

again finding that every primitive element lies on one of them.

For $(q, r) = (3, 3)$, write $F_{27} \cong \mathbb{F}_3[x]/(x^3 + 2x + 2)$. $2 + 2x$ is a primitive element; the set of primitive elements is

$$\{2 + 2x, 1 + 2x, 2x^2, 1 + x^2, 2x, 2 + x + x^2, 2 + x + 2x^2,$$

$$1 + 2x^2, 2 + 2x + 2x^2, 2 + 2x + x^2, 2x + 2x^2, x + 2x^2\}.$$

Choose hyperplanes

$$H_1 = \{2x^2 + a_1(1) + a_2(x) \mid a_1, a_2 \in \mathbb{F}_3\}$$

$$H_2 = \{2x + b_1(1) + b_2(x^2) \mid b_1, b_2 \in \mathbb{F}_3\}$$

$$H_3 = \{x^2 + 1 + c_1(x^2) + c_2(1+x) \mid c_1, c_2 \in \mathbb{F}_3\}.$$

Every primitive element is on at least one of them. $\square$

Data availability

No data was used for the research described in the article.

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